STUDYNEST | CLASS 11 PHYSICS

MARKING SCHEME AND MODEL ANSWERS

Chapter 13: Oscillations

Maximum Marks: 40 | Suggested time: 60 minutes

Use after attempting the paper. Accept equivalent correct methods.

1 mark: Correct fact, formula, unit or direction.
2 marks: Principle/formula (1); explanation, working or result (1).
3 marks: Relation (1); working/reasoning (1); final result and unit (1).
4 marks: Principle (1); setup/equation (1); working (1); result/interpretation (1).
Mark splits are a guide; award partial credit for valid steps.

Time for one cycle.

f=1/T.

F=-kx.

At equilibrium.

At extreme displacement.

sqrt(k/m).

2pisqrt(l/g).

kA^2.

2 Hz.

0.314 s.

a=-omega^2x.

2.

Zero.

Small angular displacement.

omega=20 rad/s; Tapprox 0.314 s.

0.4 m/s and 1.6 m/s^2.

approx 2.01 s.

K max/U min at equilibrium; U max/K zero at extremes; total energy constant without damping.

E=1 J; U=.25 J.

Damping transfers mechanical energy into heat/sound and reduces amplitude.